Conjecture 00000002473: rank and crank already disagree at two

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1 The necessary clause being refuted

The English statement includes the explicit assertion that "the two statistics agree outside the exceptional classes." In its stated Ramanujan-congruence context, the exceptional sizes are

$$n \equiv 4 \pmod 5, \qquad n \equiv 5 \pmod 7, \qquad n \equiv 6 \pmod{11}.$$

The integer $n = 2$ lies in none of these classes. We prove disagreement both for individual partitions and for the complete distributions, including distributions reduced modulo each of $5, 7, 11$. Thus no choice between these usual meanings of agreement repairs this clause. The accompanying Chinese text is unclear; the original bilingual source is preserved without alteration, and this counterexample addresses the clear English necessary clause. The further, unspecified near-bijective refinement cannot make a false necessary clause true and is not needed for the contradiction.

2 Standard definitions and a complete example

For a partition λ, let $L(\lambda)$ be its largest part, $\ell(\lambda)$ its number of parts, $o(\lambda)$ its number of ones, and $\mu(\lambda)$ the number of parts strictly greater than $o(\lambda)$. The usual statistics are

$$\operatorname{rank}(\lambda) = L(\lambda) - \ell(\lambda), \qquad \operatorname{crank}(\lambda) = \begin{cases} L(\lambda), & o(\lambda) = 0, \\ \mu(\lambda) - o(\lambda), & o(\lambda) > 0. \end{cases}$$

There are exactly two partitions of two, with the following values:

λ	L	ℓ	rank	crank
(2)	2	1	1	2
$(1,1)$	1	2	-1	-2

The rank distribution therefore has values $-1, 1$, once each, whereas the crank distribution has values $-2, 2$, once each. In particular, exactly one partition has rank one and none has crank one. Modulo 5, 7 or 11, exactly one partition has rank congruent to one and none has crank congruent to one.

These definitions and the three Ramanujan classes are recorded in Bringmann–Mahlburg–Rhoades, *Asymptotics for rank and crank moments*, introduction, equations (1.2)–(1.3). The familiar exceptional generating-function convention at $n = 1$ is irrelevant: this example uses actual partitions of $n = 2$.

3 Formal objects and exhaustive classification

Lean uses Mathlib's `Nat.Partition n`, whose parts are a multiset of positive natural numbers with sum n. The largest part is the maximum of the actual parts; rank subtracts the multiset cardinality in $\mathbf{Z}$. The crank counts actual ones and actual parts greater than that count, also subtracting in $\mathbf{Z}$. Negative values are not truncated by natural-number subtraction.

The definitions `two` and `oneOne` are genuine members of `Nat.Partition 2`. The theorem `partition_two_classification` proves they exhaust that type. If a part two occurs, its remaining parts have sum zero and positivity forces them to be absent. Otherwise every positive part is one; the sum forces exactly two such parts. Extensionality of the partition type identifies the two cases with the named objects. Their distinctness is proved from their different part counts. The theorem `univ_partition_two` consequently identifies the full finite universe with these two objects.

The definitions `distribution` and `residueDistribution` count filtered subsets of this full universe, for arbitrary sizes and integer statistics. The four rank/crank value theorems are computed from the definitions. The integer and modular count theorems then use the exhaustive universe equality. These are counts over all partitions, not over an assumed or incomplete test list.

4 Formal negations

The predicate `RamanujanExceptional` is exactly the union of the three size residue classes above. `two_nonexceptional` proves that two avoids all of them. The final theorems negate each of the following universal claims:

1. Outside the exceptional sizes, rank equals crank on every individual partition.
2. Outside those sizes, the full integer-valued rank and crank distributions are equal.
3. Outside those sizes, their residue distributions are equal for all three moduli $5, 7, 11$.

The explicit value-one discrepancy supplies the latter two contradictions. Nothing here contradicts the actual Ramanujan congruences or claims that rank and crank should be interchangeable in the established theory.

5 Verification

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

Source-bound receipts record a separate new Main build, sixteen transitive axiom audits and official fresh kernel replay. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed the full formal chain. This is AI review, not human certification. Organizer acceptance is pending.